Status	Finished
Started	Sunday, 25 October 2026, 10:21 PM
Completed	Sunday, 25 October 2026, 10:29 PM
Duration	8 mins 2 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 1000
c. 500
d. 100

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The middle element is chosen randomly

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the middle of the list
c. At the beginning of the list
d. In the right sublist ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By dividing the list into halves
c. By comparing each element in the list sequentially ✓
d. By sorting the list first

Question **5**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Hash
c. Linear
d. Both (1) & (3)

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Finding
c. Discovering
d. Searching ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99
b. 90 and 99 ✓
c. 89 and 94
d. 90 and 100

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n/2
c. n ✓
d. log n

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓
b. Jump search
c. Hashing
d. Binary search

Question **10**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(1) ✓
c. O👎
d. O(log n)

Question **11**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Bubble ✓

c. Insertion

d. Complexity

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Linear search ✓
c. Merge search
d. Quick search

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It can only be applied to large lists
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. Number values should only be present

Number values should only be present

c. The list must be sorted
d. There should be direct access to the middle element in any sublist

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the right sublist
c. In the left sublist ✓
d. In the entire list